

Active 3-band Equalizer Test Report

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Objective

The purpose of this test was to determine how well the system under test conformed to specifications. Additionally, it was intended to discover the viability or lack of viability of the prototype as currently designed for application in real music.

System Under Test

The system under test is a 1st prototype of an active 3-band equalizer. It employs an input buffer in series with 3 active band-pass filters in series with a summing amplifier. A block diagram of the system can be seen in figure 1 below.

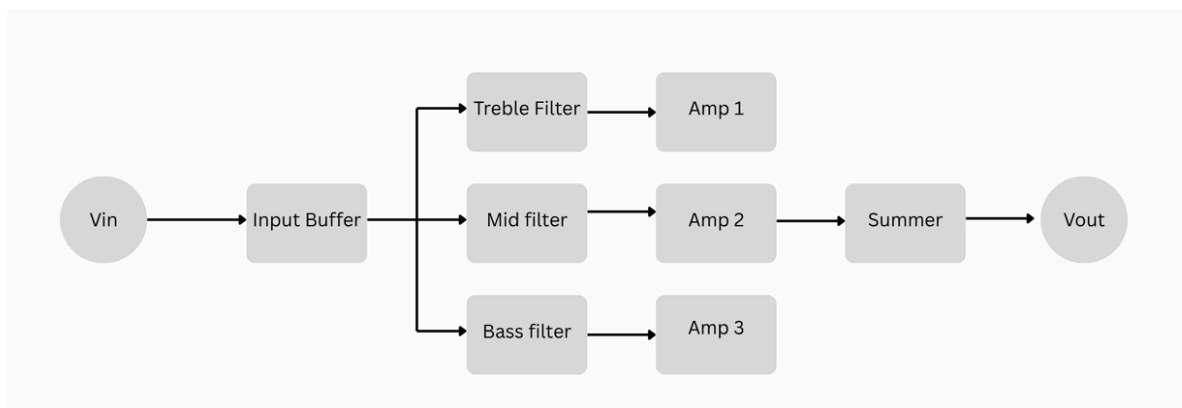


Figure 1. Block diagram of system under test

The system operates with V_{in} being a 100-500 [mVrms] sine wave. Under real conditions, the input signal is a composite of many frequencies with a periodic, heavy transient. It is powered by dual 15V rails with local decoupling employed for rail stability.

The system is tested to meet all requirements laid out in eq_spec.pdf. These requirements, while not summarized here, include requirements for gain linearity, frequency response, band independence and noise.

A schematic of the system can be seen below in figure 2.

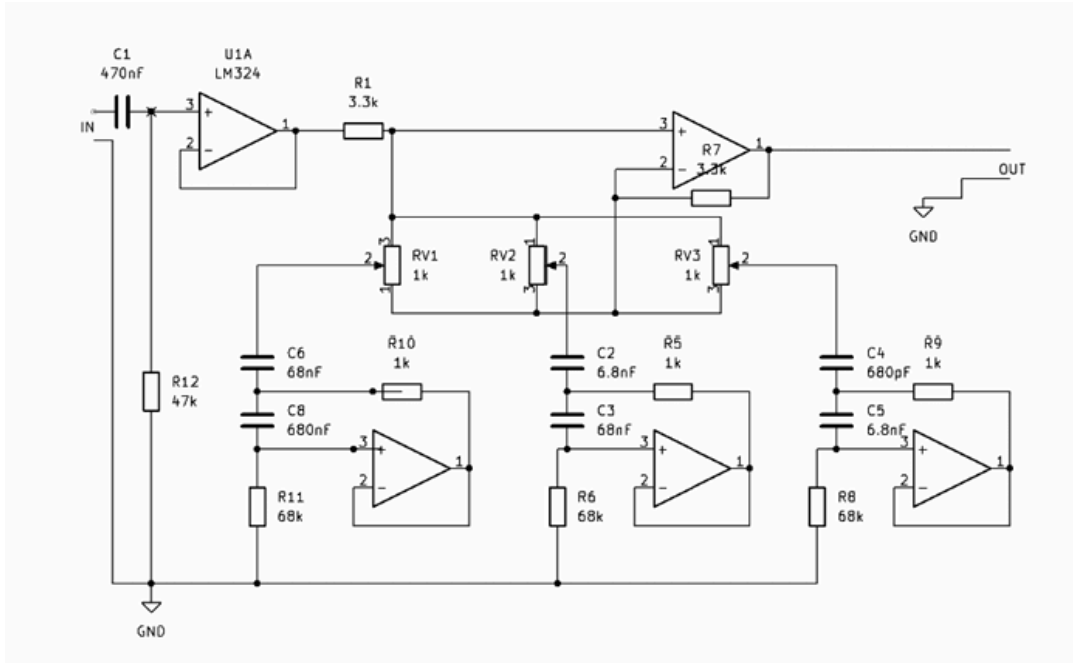


Figure 2. Schematic of system under test

Test Procedure

The test procedure is set forth in a separate document, Test plan.pdf. The procedure was designed to test that the system meets each of the requirements laid out in the specification.

In general, the test procedure proceeds in 3 phases: 1) continuity testing, 2) controlled signal testing, 3) instrument signal testing. The test procedure was followed in the lab as outlined in the test plan.

Results

The following results are laid out by requirement category.

Frequency Response

A. Data

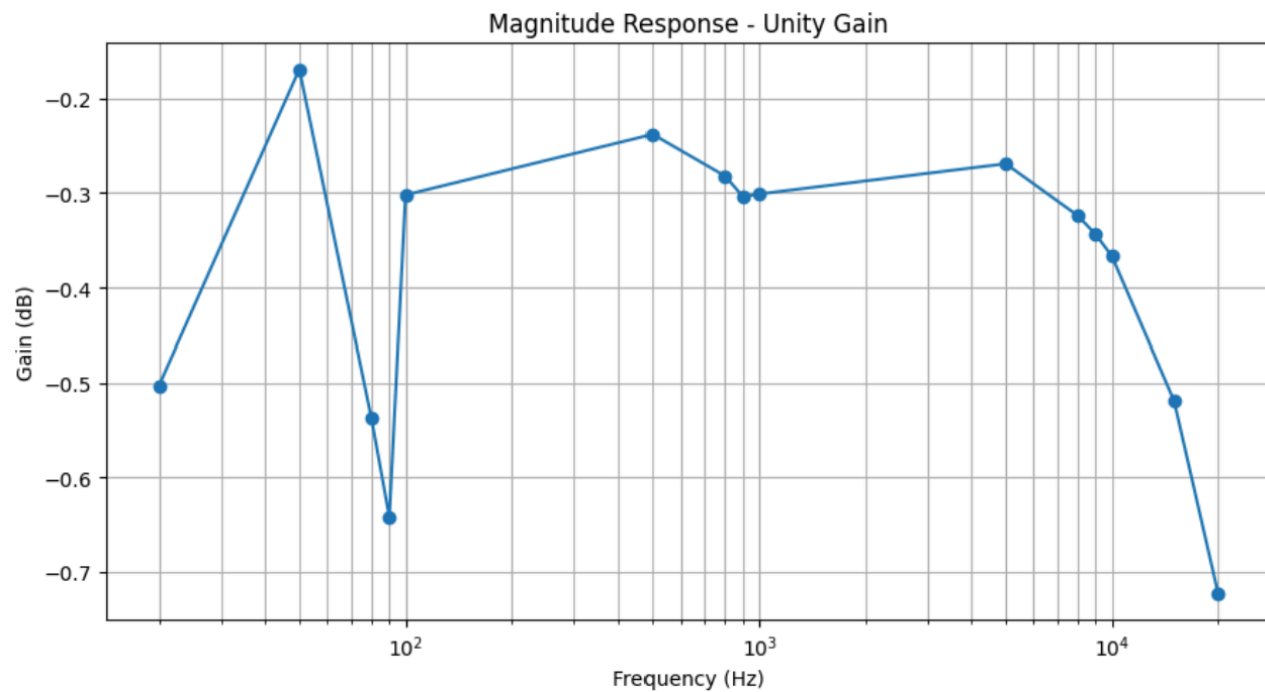


Figure 3. Frequency response at unity gain

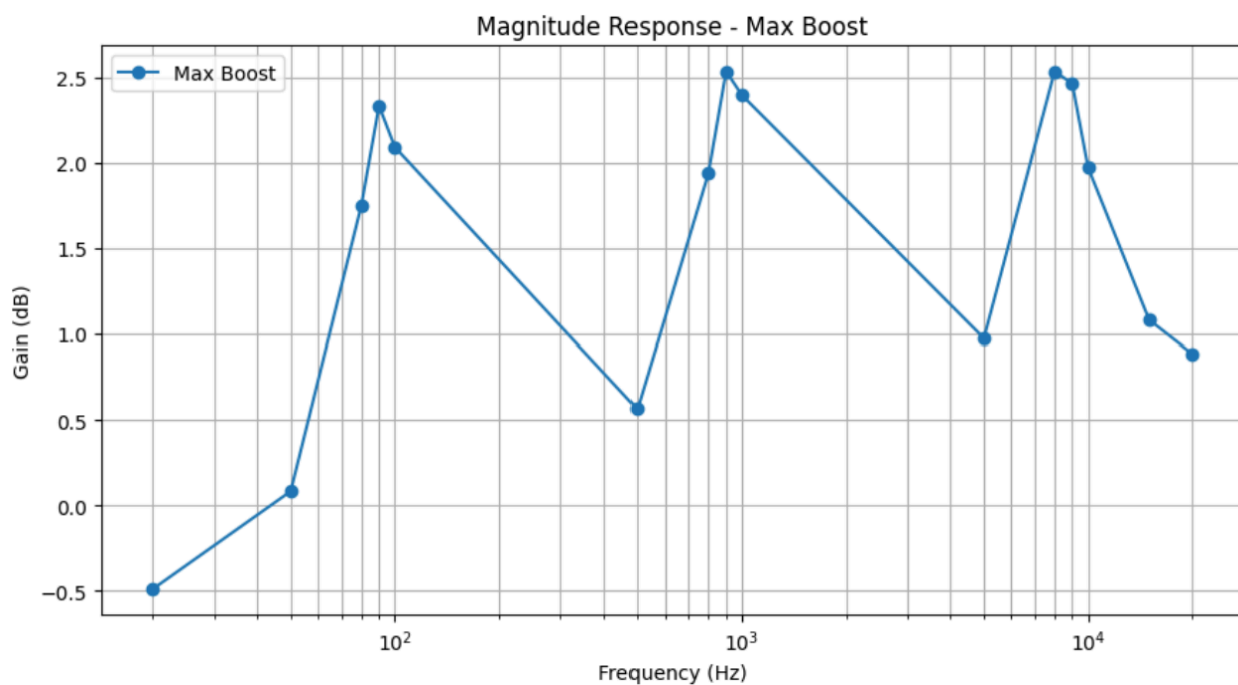


Figure 4. Frequency response at max boost (all 3 bands)

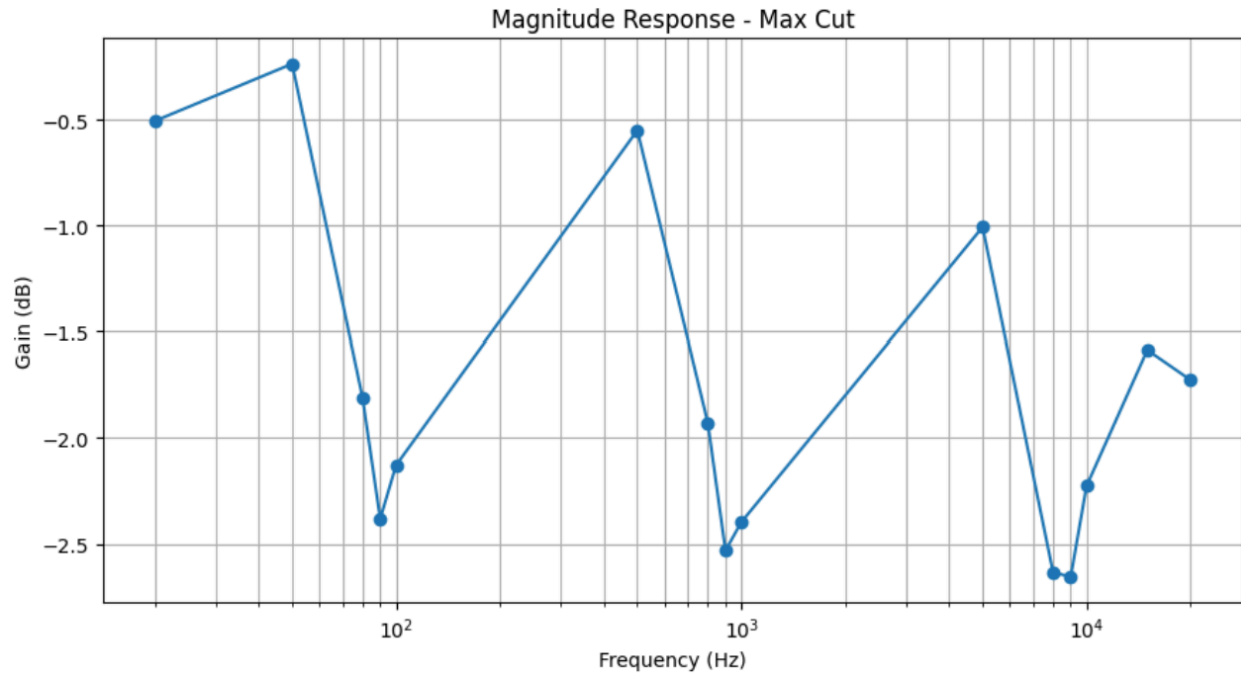


Figure 5. Frequency response at max cut (all 3 bands)

B. Relevant requirements

Requirement	Requirement Description	Pass/fail	Notes
FREQ-001	Center frequencies at 90, 900 & 9000Hz $\pm 10\%$	PASS	
FUNC-003	Nearly flat response at unity	PASS	
FREQ-002	$\pm 1dB$ max at unity	PASS	Imprecise setting of pot knob
GAIN-001	$\pm 10dB$ at f_c	FAIL	See discussion

Band Independence

Bands were all set to max cut then 1 was boosted at a time. The quantity ΔG was measured as the difference at that frequency between the baseline value and the boosted value. If operating properly, ΔG should be small for all bands besides the one being boosted at the time.

A. Data

	ΔG at 90Hz (dB)	ΔG at 900Hz (dB)	ΔG at 9000 Hz (dB)
Bass Boost	4.707	0.725	0.747
Mid Boost	0.007	4.33	0.730
Treble Boost	0.001	0.009	3.647

Table 1. Band independence data, difference from baseline

B. Relevant Requirements

Requirement	Requirement Description	Pass/fail	Notes
FUNC-002	Band independence (general)	PASS	
FREQ-003	$\Delta G < 1\text{dB}$	PASS	

Gain linearity

The gain at the center frequency of all 3 bands was measured at various input amplitudes from 20 [mVrms] to 500 [mVrms]. It was observed that gain remains linear. It is observed that gain remains approximately constant in the amplitude range tested, which corresponds to the dynamic range of a guitar.

A. Data

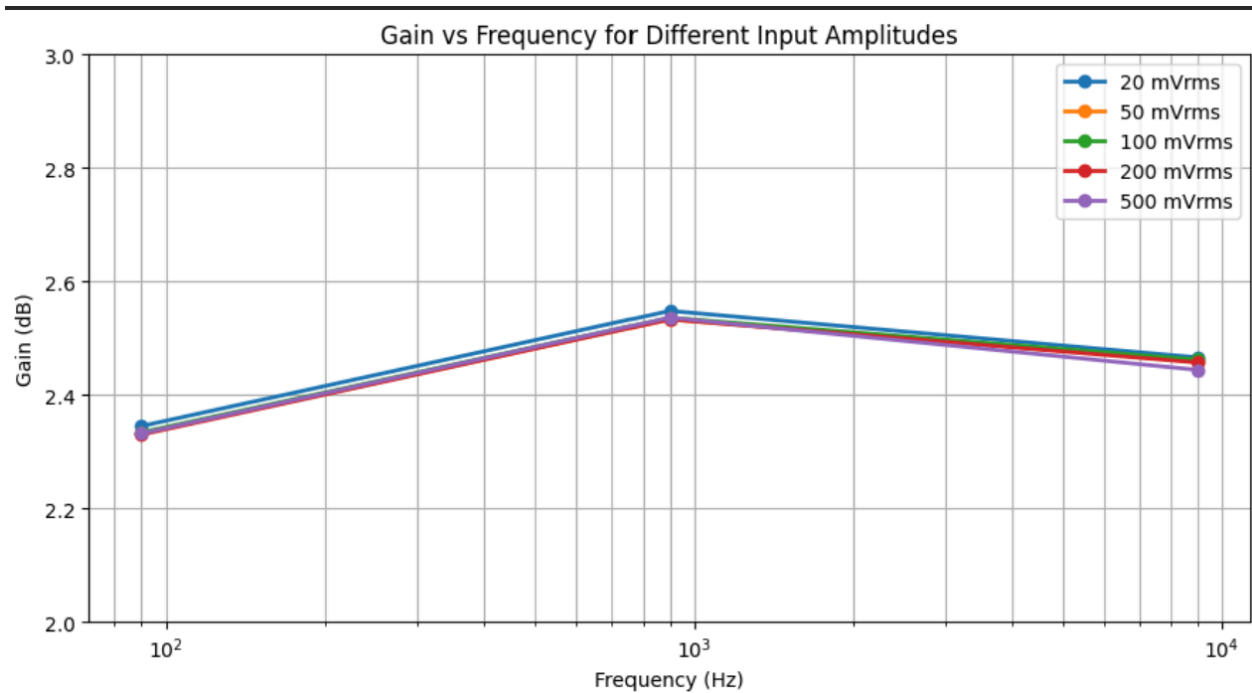


Figure 6: Gain linearity at 5 input amplitudes

B. Relevant requirements

Requirement	Requirement Description	Pass/fail	Notes
FUNC-005	Linearity in operating range	PASS	
GAIN-002	Linear gain between 20 and 500 [mVrms]	PASS	

Instrument Signal Testing

A. Data

The device was observed to make a qualitative difference in frequency content in the 3 specified frequency ranges. More conclusive, quantitative testing can be conducted at a later date.

B. Relevant Requirements

Requirement	Requirement Description	Pass/fail	Notes
FUNC-001	Audible boost/cut	PASS	

Discussion

The testing above demonstrates that all but 1 requirement are met by the device. This requirement, which specified a minimum $\pm 10dB$ gain at center frequencies, could be met in a second prototype after a simple redesign. Specifically, the feedback ratio of the summer should be increased approximately 5-fold to ensure gains reach 10dB.

The device accomplished its objective and, while not viable for immediate production, it proved to adhere well to requirements overall. Future iterations should address the low gain issue and conduct more comprehensive noise and instrument signal testing.